

For example, if you had borrowed \$10,000—for total assets of \$20,000—then a 25% loss wipes out \$5,000 of your capital, leaving you with \$15,000 at the end of the year. To add insult to injury, you must pay back the \$10,000 loan, plus the (assumed) 5% interest. At year's end, therefore, your equity (your assets minus your liabilities) has dwindled to a mere \$4,500. Your original equity investment of \$10,000 has lost 55% of its value, even though the market (fund, stock) only fell 25%. That's "the ugly" side of leverage.

So, should you leverage? Is it a good idea? Will you make money? After all, in the long run, markets go up, don't they? Well, by now you should know that there are no simple answers to these questions; all you can do is look at the odds. What are the odds that you will benefit from leverage? What are the odds that you will regret exercising it? First, let's take a good look at the upside. Suppose you are faced with a stock, mutual fund, or investment that is expected to earn 15% on average, in the long run. However, because there are no free lunches in the world, you have to contend with some risk. Assume that the volatility risk of this particular investment is 30%. So, facing this kind of investment, what happens if you leverage yourself?

You must pay particular attention to two factors: the leverage ratio and the borrowing (margin account) interest rate. Both will affect your outcome and the odds. The leverage ratio is the amount that you borrow to invest, expressed as a percentage of your original equity or capital. For example, if you have \$10,000 and borrow \$10,000, your leverage ratio is 100%. If you borrow only \$5,000, your leverage ratio is 50%. If you don't borrow at all, your leverage ratio is 0%.

Next is the interest rate factor. This is simply the after-tax rate of interest that you'll be paying on the borrowed capital. In the present climate, the rate can range anywhere from 3% to 10%, depending on many considerations. The most important of these are your credit worthiness and the odds of your going bankrupt. Clearly, you want to borrow at the lowest possible interest rate so that you get to keep as much of your market gains as possible.

When you put the two factors together—leverage ratio and interest costs—you can do some intelligent analysis of how the leverage will affect your investments. The numbers in Tables 4.2 through 4.4 come from a Monte Carlo computer simulation. Monte Carlo computer